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INTRODUCTION TO FINTECH

Case Study 01 – Python-Based FinTech Analytics
Group 6 - Dataset 1

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Contents

1. Introduction.....	3
2. Data Understanding and Preparation.....	3
3. Exploratory Data Analysis.....	5
4. Modelling Methodology.....	6
5. Model Evaluation and Results.....	7
6. Model Interpretation and Risk Segmentation.....	8
7. Fairness, Limitations and Practical Considerations.....	8
8. Conclusion.....	10

Credit Risk Prediction Using Machine Learning

1. Introduction

Credit risk is one of the main challenges faced by financial institutions when providing loans. A bank or lending institution needs to decide whether an applicant is likely to repay a loan or become a defaulter. Making this decision only through traditional methods can be difficult because many factors, such as credit history, income, debt level, employment status, and previous defaults, can affect repayment behaviour.

Machine learning can support this process by identifying patterns in historical loan data and using them to estimate the risk of future default. However, a useful credit risk model should not only achieve good predictive performance. The data must first be cleaned carefully, the model must be evaluated using appropriate measures, and issues such as fairness, data leakage, and practical business costs must also be considered.

This study develops and evaluates machine learning models to predict loan default using `dataset01.csv`. The target variable is `default_status`, where 1 represents a default and 0 represents a non-default. The study also examines the main factors associated with default and considers how the model could be used as a decision-support tool.

2. Data Understanding and Preparation

The original dataset contained **4,715 observations and 23 variables**. The initial data quality assessment identified several problems, including duplicate records, missing values, inconsistent categorical values, different date formats, and unrealistic numerical values.

A total of **115 exact duplicate rows** were identified and removed. After removing these duplicates, the final dataset contained **4,600 unique applicant records**. The default rate was

approximately **28.3%**, meaning that the dataset was moderately imbalanced because non-defaulters were more common than defaulters.

Categorical variables such as gender, marital status, education, employment status, home ownership, and previous default status were standardized. Differences in spaces, capitalization, underscores, and binary formats such as Y/1 and N/0 were corrected.

Numerical values were also checked for unrealistic observations. Ages below 18 or above 100, employment length recorded as 999, and credit scores outside the normal range of 300–850 were treated as missing values rather than being used directly. The `application_date` variable contained three different date formats, so the dates were interpreted using year-month-day, day-month-year, and month-day-year patterns. This interpretation is a limitation because the original date-format definition was not clearly documented.

Missing values were handled inside the modelling pipeline after splitting the data. This was important because it prevented information from the test set from influencing the training process.

Customer ID and application date were not used as predictive variables. Gender and marital status were also excluded from prediction but retained for later fairness analysis. Age was retained because it may have a relationship with credit risk. After preprocessing, **18 initial predictor variables** were considered for modelling.

The data was divided using an **80:20 stratified train-test split**. This produced 3,680 training observations and 920 test observations while maintaining a similar default rate in both groups.

3. Exploratory Data Analysis

The exploratory analysis showed several meaningful relationships between applicant characteristics and loan default.

Credit score was the clearest factor associated with default. Its correlation with default was approximately **-0.28**, indicating that applicants with higher credit scores generally had lower default risk. The observed default rate was around **43% among applicants with credit scores between 300 and 500**, compared with approximately **13% among applicants with scores between 701 and 850**.

Previous default history was another important factor. Applicants with a previous default had an observed default rate of approximately **45%**, compared with around **25% among applicants without a previous default**.

Employment status also showed differences. The default rate among unemployed applicants was approximately **38%**, while other employment groups were generally around **27–28%**. Higher debt-to-income ratios were also associated with somewhat higher default risk.

In contrast, loan purpose, home ownership, and loan term showed relatively small differences in default rates. The `interest_rate_pct` variable had almost no correlation with default and provided very little additional predictive value. It was therefore removed from the final feature set. This was also considered important because interest rates may already be assigned based on a lender's assessment of risk, which could create a circular relationship in the model.

Overall, the EDA suggested that credit history, previous default behaviour, employment conditions, and debt burden were more useful for predicting default than several basic loan characteristics.

4. Modelling Methodology

Three machine learning approaches were evaluated:

1. Logistic Regression
2. Random Forest
3. Gradient Boosting

A simple dummy classifier was also used as a baseline. This model always predicted the majority class, meaning that it predicted no default for every applicant. Although it achieved approximately **72% accuracy**, its ROC-AUC was only **0.500** and its recall for actual defaulters was zero. This demonstrated why accuracy alone is not an appropriate measure for credit-risk prediction.

The three machine learning models were evaluated using **5-fold stratified cross-validation**. Initial ROC-AUC values were approximately **0.685 for Logistic Regression, 0.679 for Random Forest, and 0.678 for Gradient Boosting**.

After tuning, the ROC-AUC values increased slightly to approximately **0.687 for Logistic Regression, 0.687 for Random Forest, and 0.691 for Gradient Boosting**. Since the difference between the models was very small, Logistic Regression was selected based on the predefined rule of choosing the simpler and more interpretable model when performance was within 0.005 ROC-AUC of the best model.

Class weighting was also examined because missing a genuine defaulter can be costly. Although class weighting had little effect on ROC-AUC, it substantially improved recall for defaulters. Therefore, the balanced Logistic Regression model was retained as the final model.

5. Model Evaluation and Results

The final model was evaluated on the test dataset, which was kept separate from model development. The locked test-set **ROC-AUC was 0.715**, with a bootstrap 95% confidence interval of approximately **0.678–0.750**. The corresponding Gini coefficient was **0.430**, while the PR-AUC was approximately **0.463**.

The model's ROC-AUC was only slightly higher than the baseline Logistic Regression result of approximately 0.711. This indicates that model tuning produced only a modest improvement. However, the final model still provided useful separation between higher- and lower-risk applicants.

A probability threshold of 0.50 was not automatically used. Instead, the threshold was selected using out-of-fold training predictions. A **3:1 false-negative to false-positive cost ratio** was assumed because failing to identify a genuine defaulter was considered more costly than incorrectly flagging a non-defaulter. Based on this assumption, a threshold of **0.41** was selected.

At this threshold, approximately **60.3% of applicants were flagged as high risk**. The model achieved approximately **83.5% recall**, meaning it identified most of the actual defaulters. However, precision was approximately **39.1%**, while the false-positive rate was approximately **51.2%**.

These results show an important practical trade-off. The model can identify a large proportion of potential defaulters, but it also flags many applicants who would not actually default. Therefore, the model should be used mainly for **risk screening and prioritisation**, rather than automatically rejecting loan applications.

6. Model Interpretation and Risk Segmentation

One advantage of Logistic Regression is that its results can be interpreted relatively easily. Credit score was the strongest model input, with a coefficient of approximately **-0.619**. This negative coefficient indicates that higher credit scores were associated with lower predicted default risk.

Previous default history was also important. Unemployment and higher debt-to-income ratios were associated with increased predicted risk.

Permutation importance provided additional evidence. Shuffling credit score reduced the test ROC-AUC by approximately **0.17**, while shuffling previous default reduced it by around **0.03** and debt-to-income ratio by approximately **0.01**. This suggests that credit score was considerably more influential than the other examined variables.

The model was also used to create risk groups. The observed default rate increased from approximately **9.2% in the lowest-risk group to 50.0% in the highest-risk group**. The highest-risk group contained approximately **35.4% of all observed defaults**.

These risk groups could support practical lending decisions. For example, lower-risk applicants could follow the normal assessment process, medium-risk applicants could receive additional checks, and higher-risk applicants could receive more detailed review. However, actual cut-off points should be determined using the institution's financial costs, risk appetite, lending policies, and customer considerations.

7. Fairness, Limitations and Practical Considerations

Several limitations should be considered before using the model in a real lending environment.

First, the dataset appears to be synthetic or simplified, and the default rate of 28.3% may not represent a real financial institution's portfolio. The definition and observation period for default are also not clearly documented.

Second, the dataset contains some potentially contradictory information, such as unusual combinations of age and employment length. In addition, the study uses a random train-test split, which may not fully represent how a model performs when predicting future customers.

An out-of-time evaluation was therefore considered. When the model was trained using 2021–2023 data and tested on 2024 data, the ROC-AUC was approximately **0.693**, compared with approximately 0.715 for the random test split. This difference suggests that model performance may change over time.

Fairness was also considered across gender, marital status, age groups, region, and application year. Gender and marital status were excluded from prediction, but removing these variables does not completely eliminate possible bias because other variables may act as proxies. Small groups, particularly some gender and marital-status categories, also produced less stable estimates.

Finally, the 3:1 cost ratio used for threshold selection was an assumption rather than a real financial cost estimate. In a real organisation, threshold selection should consider actual losses, recovery rates, profitability, customer value, and risk appetite.

8. Conclusion

This study developed a machine learning approach for predicting loan default using historical applicant information. After cleaning and preparing the data, Logistic Regression, Random Forest, and Gradient Boosting models were compared. Their predictive performance was relatively similar, so Logistic Regression was selected because it provided a good balance between performance and interpretability.

The final model achieved a test ROC-AUC of approximately **0.715** and showed useful separation between different levels of credit risk. Credit score, previous default history, employment status, and debt-to-income ratio were among the most important factors.

However, the model should not be considered a fully automated lending decision system. Its relatively high recall was achieved at the cost of many false positives. Therefore, the most suitable role for the model is as a **decision-support and risk-ranking tool**, where its predictions are combined with human review and existing lending procedures.

For future implementation, the model should be tested using larger and more representative real-world datasets, validated using longer out-of-time periods, calibrated if predicted probabilities are used as actual probabilities of default, and continuously monitored for performance, data drift, and fairness. Actual business cost information should also be used to select the final decision threshold.

Overall, the study demonstrates that machine learning can provide useful support for credit-risk assessment, while careful data preparation, transparent modelling, fairness analysis, and human oversight remain essential for responsible use.